

Minor Course B-Tech CSc
Department of Information Technology
w.e.f Academic Year 2025-26
Semester – 1 (2025-26)

Course Code: 25MT2U101		Course Title:Discrete Mathematic	
Scheme of Instruction		Scheme of Evaluation	
Total Duration	45 Hrs.	Max. Marks	100
Hours/Week	3 Hrs.	Continuous Internal Assessment	45
Credits	3	Attendance	05
Instruction Delivery	2-1-0	Seminar/ End Exam	50
Course Type	Minor	Exam Duration	2.30 Hr.
Instruction Mode	Regular	Evaluation	Grade

Course Objectives	- To introduce students to understand, explain and apply the foundational mathematical concepts at the core of computer science. - To understand use of set, function and relation model with practical examples, and interpret the associated operation and terminologies in context. - To learn the fundamental counting principle, permutation and combinations.
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Course Outcomes	- Formulate the problems precisely and solve real world problems logically using sets and propositions.. - Calculate number of possible outcomes using permutation, combination and find probability. - Applying graph theory concepts in structural models to arrange various objects or technology leads to new inventions and modifications.
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Syllabus Content		
Unit-1	Module-1	Rule of Sum, Rule of Product, Introduction to Counting Problems, Simple Applications in Daily Scenarios.
	Module-2	Permutations (with and without repetition), Combinations (with and without repetition), Differences and Use Cases, Applications in Selection and Arrangement Problems.
	Module-3	Algorithms to Generate All Permutations (e.g., Recursive, Lexicographic), Algorithms to Generate All Combinations,

		Pseudocode and Complexity Analysis, Applications in Computer Science (e.g., Testing, Optimization).
	Module-4	Introduction to Probability, Discrete Probability, Conditional Probability, Applications in Dice, Cards, and Real-Life Scenarios.
	Module-5	Bayes' Theorem and Its Applications, Binomial Theorem (in Counting and Probability), Introduction to Information Theory,
Unit-2	Module-1	Relational Model for Databases, Binary Relations, Properties of Binary Relations: Reflexive, Symmetric, Anti-symmetric, Transitive, Representation of Relations (matrices, digraphs).
	Module-2	Closure of Relations (Reflexive, Symmetric, Transitive Closures), Warshall's Algorithm (for Transitive Closure), Equivalence Relations, Partitions and Equivalence Classes,
	Module-3	Partial Ordering Relations, Hasse Diagrams, Lattices (Definition and Examples), Chains and Antichains, Compatible Relations.
	Module-4	Functions: Definitions and Notation, types of Functions: One-to-One, Onto, Bijective, Composition of Functions, Invertible Functions,
	Module-5	Pigeonhole Principle (Basic and Generalized), Recursive Functions: Definition and Examples, Applications in Problem Solving and Theoretical Computer Science.
Unit-3	Module-1	Basic Terminology (vertices, edges, degree, etc.), Types of Graphs: Multigraphs, Weighted Graphs, Representation of Graphs: Adjacency Matrix, Adjacency List, Incidence Matrix.
	Module-2	Subgraphs, Isomorphic Graphs, Special Graphs: Complete Graphs,

		RegularGraphs, Bipartite Graphs.
	Module-3	Operations on Graphs (union, intersection, complement, etc.), Paths and Circuits, Graph Traversals: Depth-First Search (DFS), Breadth-First Search (BFS).
	Module-4	Eulerian Paths and Circuits, Hamiltonian Paths and Circuits, Shortest Path in Weighted Graphs:Dijkstra's Algorithm.
	Module-5	Factors of a Graph, PlanarGraphs, Graph Coloring, Traveling Salesman Problem (TSP).
Text Books	1.Kenneth H. Rosen, <i>Discrete Mathematics and its Applications</i>, Tata McGraw-Hill, ISBN 978-0-07-288008-3, 7th Edition. 2.C. L. Liu, <i>Elements of Discrete Mathematics</i>, TMH, ISBN 10: 0-07-066913-9	
References	1. Bernard Kolman, Robert C. Busby and Sharon Ross, <i>Discrete Mathematical Structures</i>, Prentice-Hall of India / Pearson, ISBN: 0132078457, 9780132078450. 2. N. Biggs, <i>Discrete Mathematics</i>, 3rd Edition, Oxford University Press, ISBN 0 – 19 850717 – 8. 3. NarsinghDeo, <i>Graph with application to Engineering and Computer Science</i>, Prentice Hall of India, 1990, 0 – 87692 – 145 – 4. 4. Dr. K. D. Joshi, <i>Foundations of Discrete Mathematics</i>, New Age International Limited, Publishers, January 1996, ISBN: 8122408265, 9788122408263. 5. C. D. Cantrell, <i>Modern Mathematical Methods for Engineers</i>, Cambridge University Press, ISBN: 0521670497. 6. Eric Gossett, <i>Discrete Mathematical Structures with Proofs</i>, Wiley India Ltd., ISBN: 978-81-265-2758-8. 1. Sriram P & Steven S., <i>Computational Discrete Mathematics</i>, Cambridge	

	University Press, ISBN 13: 978-0-521-73311-3.
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Note: Tutorials based on complete syllabus of the course.